# Energy consumption estimates

(Dated: January 22, 2026)

## I. INTRODUCTION

## II. SIMULATION RESULTS

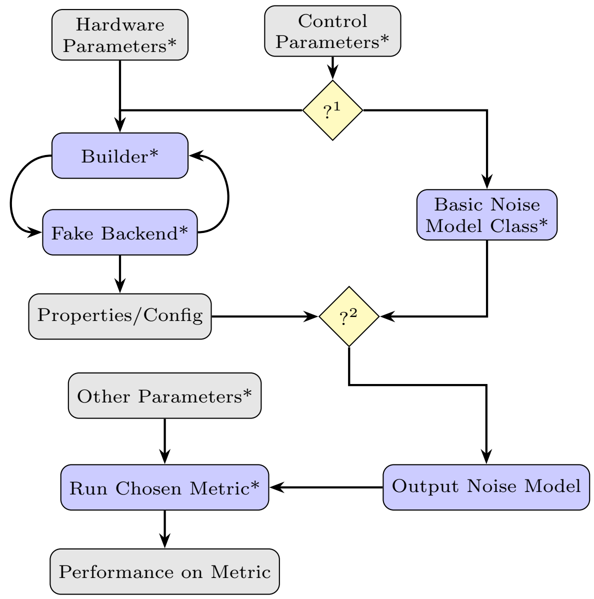
To complement this mathematical approach, we also computed simulations tracking performance on quantum algorithms as we varied temperature as the single control parameter. We computed simulations of both:

- Full quantum circuits, tracking full density matrix representations, allowing us to employ full Qiskit noise models
- Circuits using only Clifford gates, tracking qubit state using stabiliser states and only employing simpler Pauli noise models. This allowed us to simulate much larger numbers of qubits.

### A. Design Architecture

Something to be emphasised is the total modularity of the system we designed to do this. All parameters can be altered and all models can be swapped in and out, providing complete flexibility over how the performance value is computed.

Here is an overview of the system architecture:




$?^1$: Choose between an IBM Fake Backend or a basic noise model class

$?^2$: A Pauli noise model is generated if the chosen metric is a Clifford metric. If a full noise model is generated, Pauli twirling can optionally be turned off.

FIG. 1. Dataflow for simulated performance on a quantum algorithm, from control parameters to noise model to performance. All nodes noted with a * are modular and customisable.

### B. Case Studies

#### 1. *Case Study 1:* ***Quantum Volume on Legacy Backends***

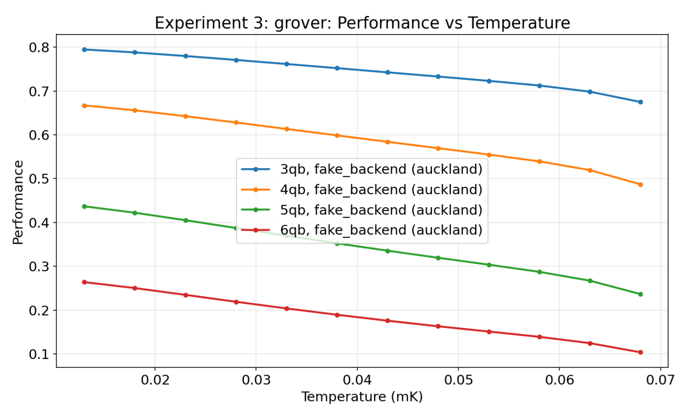



FIG. 2. Performance on Grover's algorithm using Auckland Fake Backend

These are both V2, Falcon R5 backends released

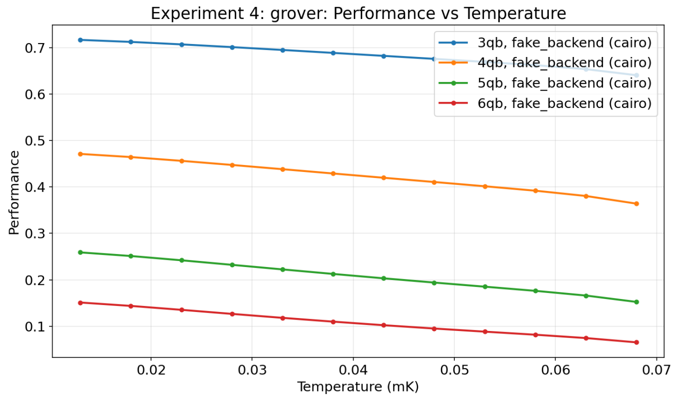



FIG. 3. Performance on Grover's algorithm using CairoV2 Fake Backend

around late 2021. You can see a stable drop in performance with increased qubit counts.

### C. Result Validity

The validity of these results hinges primarily upon the correctness of the code used to generate them. Thus, we have generated an extensive suite of unit tests to ensure the framework is operating as expected.

Moreover, the validity and applicability of these results further hinges upon the use of correct and properly catalogued assumptions used to generate them. We employ the following to mitigate the risks of compromising these qualities:

- The use of a variety of different methods of producing results, including a variety of different metrics and methods of generating noise models. These methods vary in complexity and specificity, avoiding how many assumptions are universally relied upon.
- Comprehensively cataloguing the assumptions used at every stage of generating the results, including how they change for each method of producing results.

To do this, we treat each set of results as a case study. Because there is so much overlap between the assumptions of each case study, we include a set of all set of assumptions in the appendix, and specify which ones apply for each.

There are a few assumptions that due to the inherent limitations of the simulation architecture universally apply, and are worth pointing out separately:

- All noise models are inherently gate-local (and thus also qubit-local)
- All noise models are state independent and unchanging over time throughout the whole metric simulation, and every shot and trial of the metric
- All noise models are Markovian

#### 1. *Case Study 1:* ***Quantum Volume, Simple Arrhenius Noise Model***

##### 2. *Hardware Parameters*

I make the following modelling assumptions for the IBM processors:

We use the following hardware parameters in the temperature dependent model:

- $\gamma_0$ is fit at runtime to the known values of $T_1$ at the default operational mixing chamber temperature
  - Assume the operational cryostat temperature to be 0.013K
- $\gamma_{\varphi-\mathrm{base}}$ is fit at runtime to the known values of $T_2$ at the default operational mixing chamber temperature
  - Again, assume the operational cryostat temperature to be 0.013K
- $\frac{\gamma_{MXC}}{\gamma_0}$ is 0.85.
  - This represents the maximum value from the experimental dataset reported in Ref. [1].
- $T_{env}$ as 30mK
  - Effective temperatures as low as 20mK have been observed Ref. [2].
- $\Delta$ (superconducting energy gap) of $2.88e^{-23}$
  - This is the known value for Aluminium, which we can assume the electrodes used in IBM transmon qubits are made of
- $\omega_{ge}$ is extracted from known configuration data for each backend individually for each qubit
- $N_e$, the number of effective channels in the Josephson Junction, is estimated at $1.29 \times 10^5$. We calculate this value using:
  - An estimated normal resistance of the Josephson Junction of 5k$\Omega$
  - Transparency of the junction in the subgap regime of $5 \times 10^{-5}$
- $E_c$, charging energy of the capacitors in the transmon qubits, of 0.95$\mu$eV
- $E_j$, josephson energy of the transmon qubits, of 62.1$\mu$eV

Starting with an estimated value of $\gamma_0$: $5e3$, and an estimated standard operational cryostat temperature of 0.013K, we minimise the squared error between the calculated and known, default relaxation time of the qubits. We then do the same for the value of $\gamma_{\varphi-\mathrm{base}}$ keeping $\gamma_0$ constant

The final scale factors for the value of $T_1$ and $T_2$ for each qubit are stored and applied to the calculated values to maintain granularity

When calculating gate errors, the difference between the known gate error and calculated gate error at the operational temperature is stored and assumed to be representative of the non-thermal error, and thus independent of temperature. It is then added to future calculated gate errors

[1] D. S. Lvov, S. A. Lemziakov, E. Ankerhold, J. T. Peltonen, and J. P. Pekola, Physical Review Applied **23**, 054079 (2025).

[2] S. Simbierowicz, M. Borrelli, V. Monarkha, V. Nuutinen, and R. E. Lake, Physical Review X Quantum **5**, 030302 (2024).